

Research papers

Spatio-temporal cross-validation to predict pluvial flood events in the Metropolitan City of Venice

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ABSTRACT

Due to a combination of climate change and urbanization, the instances of pluvial flooding are expected to increase in the next decades posing raising threats to properties, people and productive assets. Predicting and mapping pluvial flood-prone areas is becoming a crucial step in flood mitigation and early warnings, as well as climate change adaptation strategies, to be incorporate in urban planning. Most commonly applied machine learning (ML) procedures for pluvial flood risk assessment, neglect to account for spatio-temporal constraints, leading to overoptimistic models that underestimate the prediction error. In this paper, we propose a novel ML-based methodology for pluvial flood risk prediction in the Metropolitan City of Venice which, introducing a features selection process and spatio-temporal cross-validation, permits to reduce overfitting of the resulting ML models. Spatio-temporal characteristics of floods are derived from a dataset of 60 historical events occurred in the area between 1995 and 2020. Logistic Regression (LR), Neural Networks (NN) and Random Forest (RF) models are applied to identify and prioritize sub-areas that are more likely to be affected by pluvial flood risk, considering the daily precipitation amount and 12 different triggering factors. The models were validated using Random Cross-Validation (R-CV) and Leave Location and Time Out cross-validation (LLTO-CV), that split data in training and validation set considering both time and space. In addition, a forward features selection procedure was applied to identify the features, among the triggering factors, that better face spatio-temporal overfitting in pluvial flood prediction based on the Area Under the Curve (AUC) score. Results suggest that Logistic Regression and LLTO-CV represent the most reliable model to predict pluvial flood events in new spatio-temporal conditions, while, among the triggering factors, distance to river and distance to road resulted the prominent ones.

1. Introduction

Among the observed natural and anthropogenic calamities, water-related disasters (i.e., floods, droughts) are undoubtedly the most recurrent ones, posing increasing threats to human life, assets and activities (CRED, 2018). According with the United Nation World Water Development Report (UNESCO 2020), the 74% of disasters registered worldwide in the last 20 years were water-related with a total number of deaths caused only by floods and droughts exceeding 166,000 and a total economic damage of almost US\$700 billion.

Despite coastal and river flooding usually receive more attention as they are generally larger and longer-lasting events, recent research suggested that, due to the frequent nature of pluvial floods, cumulative direct damages of such type of flood equals or may even exceed those

derived from riverine and coastal ones (EEA 2020; Nicklin et al. 2019). Pluvial flood distinguishes it from riverine and coastal flooding. River flood are characterized by gradual riverbank overflows caused by extensive rainfall over an extended period, while coastal flooding occurs as a consequence of extreme tidal conditions caused by severe weather (e.g., storm surge, strong winds) (Maddox, 2014). By contrast, pluvial floods occur when intense rainfall cannot drain quickly enough by urban drainage system (Kalantari et al., 2019; Sperotto et al., 2015). They are usually associated with high intensity rainfall rates, but can also occur with low rainfall rates, when the ground has low permeability and is urbanized (Falconer et al., 2008). The intensification of heavy rainfall phenomena due to climate change (IPCC, 2021) accompanied by urbanization, land-use/land-cover change, modification in river morphology, agricultural practices and soil degradation is likely to

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exacerbate pluvial flood risks (Hoegh-Guldberg et al., 2018; Shahabi et al., 2021), especially in urban areas where the formation of micro-climates can further influence rainfall patterns (Kalantari et al., 2019; Shahabi et al., 2021). Being able to identify and predict pluvial flood-prone areas at local level and how such areas will be shaped by future rainfall patterns is crucial to ensure sufficient time for people to respond to future floods, as well as to set the basis for the prioritization of flood risk reduction investments. Accordingly, pluvial flood risk mapping is becoming a key step in the development of flood mitigation and early warnings, as well as climate change adaptation strategies, informing the implementation of the Sendai Framework for DRR 2015–2030 (UNISDR 2015) and the multi-hazard risk assessments compelled by the EU Civil Protection Mechanism. According to the existing literature (Maragno et al. 2021), multiple and diverse methodologies have been developed to study the pluvial flood risk in urban areas. The most widely applied method for flood mapping involved the use of hydraulic and hydrodynamic models (Li, 2011) to simulate the inundated areas and water depth based on historical or predicted rainfall data. Different kind of models have been employed so far including, among others, MIKE FLOOD (Löwe et al., 2017; Kaspersen et al., 2017), MIKE – 21 (Wen et al., 2013; Rana et al., 2012; Shrestha et al., 2020), MIKE – 3 (Bolanos et al., 2014; Sørensen et al., 2004; Pietrzak et al., 2002), HEC-RAS (Shrestha et al., 2020; Ranaee et al., 2009; Thakur et al., 2017; Vojtek et al., 2019) Soil Water Assessment Tool (SWAT) (Pagliero et al. 2014; Yoon et al. 2021), Info Works ICM (Cheng et al. 2017), HYDROTEL (Ben Aissa et al., 2012). However, despite hydraulic and hydrodynamic flooding models are characterized by a high level of accuracy their practical application for pluvial flood assessment is often hampered by the limited understanding of hydrological and hydrodynamic processes, by complex input data requirements, the high computational time and costs (Yan et al., 2021; Wagenaar et al., 2020; Beyer and Laney, 2012).

These kinds of models, require a high amount of data and variables for their calibration to be able to accurately represent the flood dynamic in the urban environment characterized by highly complex flow paths and microscale topographic features dramatically increasing the computational burden (Samela et al., 2020).

Physical based-models can simulate water flows at sub-daily level and at multiple timescales (e.g., hourly rate, three- or six-hourly rate), however it expands the computational resource demand, which lead to inconsistent predictions when setups overlap in the time ranges (Boithias et al., 2017; Gauch et al., 2021). Moreover, at present, hydraulic and hydrodynamic models require a long calculation time, problem which become prominent when working at the urban scale requiring high spatial resolution. It represents a bottleneck of rapid pluvial flood mapping and especially for those application that require re-iterative analysis (Guo et al. 2020) or prediction over alternative precipitation scenarios (Leitao et al. 2010; Ke et al., 2020).

For such reason, the use of process models may not always be feasible for real-time flood mapping and operative flood warning to be used for urban flood risk assessment and emergency management when, instead, accurate and timely outputs become more important than system understanding (Wagenaar et al., 2020).

On the other hand, Geographic Information System (GIS) and Remote Sensing techniques have been developed as useful tools for spatial analysis in flood hazard assessment (Ghorbanzadeh et al., 2019a). Multi-Criteria Decision Analysis (MCDA) functions are commonly employed to develop indices and ranking alternatives starting from geographical indicators. It is hence widely used in GIS-based models. However, a limiting factor of the approach is its high dependence on experts' knowledge, which increases the uncertainty while assessing the results (Chowdary et al. 2013). Khosravi et al. (2019) reviewed and compared different MCDM methods (e.g., TOPSIS, AHP, ANP, SAW) used for mapping flood susceptibility. Respect to other data driven methods (i.e. ML) they present some weakness. Firstly, they are data highly intensive in recording decision-makers preferences, secondly, the weighting system for the variables requires strong

assumptions from decision-makers which could make them subjective (Velasquez and Hester, 2013; Khosravi et al., 2019). These are critical aspects to consider when selecting suitable methods for flood susceptibility modeling.

These intrinsic limitations of the traditional flood models have inspired the research community to explore and test new methodologies that leverage these challenges, among these, the ML models.

In the last years, the use of Machine Learning (ML) methods to predict natural hazard risk has increased exponentially. ML models are increasingly used as modeling tool because of their high performance and predictive capability (Ahmadlou et al. 2019). ML embraces multiple disciplines, such as informatic, statistics, environmental science, mathematics permitting very diverse sources of data to be integrated to solve complex environmental problems. Nevertheless, its increasing popularity stem from the fact that it does not need complex mathematical expression to describe and predict physical processes (Mosavi et al., 2018; Shahabi et al., 2021). ML has been involved in the analysis of different natural hazards typologies including floods (Chen et al. 2020; Dodangeh et al. 2020; Mahato et al. 2021; Nandi et al. 2016), landslides (Costanzo et al. 2014; Pradhan 2013; Tien Bui et al. 2012; Bui et al., 2016), and wildfires (Dutta et al., 2016; Fiorucci et al., 2019; Ghorbanzadeh et al., 2019b; Jaafari et al., 2019).

Wagenaar et al. (2020) claim that ML could have the potential to change the way flood assessment is performed thanks to the accuracy improvements and the reduction of calculating time and model development cost. Thanks to their high performances in processing big datasets with enough iteration, detail and flexibility (Oxford Sustainable Finance Programme, 2018) ML is able to frame simpler and less computational models respect to traditional flood models opening their applicability a large range of contexts.

Specifically, ML present relative short computational time respect to physically based-models making them particularly suitable for real-time flood mapping in early warning systems and for short-term flood forecasting supporting emergency response and preparation phases (Bonafilia et al., 2020; Wagenaar et al., 2020).

In addition, ML models permit to jointly compute different time-scales in a hierarchical process, by integrating forcings of different temporal resolutions and generating accurate and consistent predictions. These models require far smaller computations than individual physical flood models per timescale (Gauch et al., 2021). According to Khosravi et al. (2019) ML methods outperform the MCDM techniques because of their greater accuracy and higher prediction performance. Finally, thanks for their flexibility in integrating heterogeneous source of data, they can benefit from the integration in a real-time setting of data from emerging technologies including satellite imagery, aerial imagery, crowdsourcing, social media, and mobile GPS and Call Data Records (CDR) (Lee et al., 2012).

According with Zennaro et al. (2021), considering also climate change impacts, ML can substantially improve current risk assessment procedures, permitting to capture the complex interactions and causal nexus occurring among the variables that may alter the equilibrium of exposed systems affected by natural and anthropogenic drivers, especially in the assessment of flood events that require deep exploration of the complex and site-dependent mechanisms.

In the frame of climate change risk assessment, ML algorithms can be used to link together the components of the risk (i.e., hazard, vulnerability and exposure). Specifically, in the hazard appraisal, ML has been adopted for instance for short term flood and flash flood forecasts and for monitoring phenomena in the long term under different climate change scenarios (Chen et al., 2019; Kim et al., 2015). While, ML has supported the vulnerability component to predict human-related vulnerabilities (e.g., demographic and socio-economic conditions) and for climate-related vulnerability (Buckland et al., 2019; Saha and Pal, 2019). The exposure component mainly concerns the detection and prediction of exposed targets in time under changing climatic and land-use scenarios (Karan and Samadder, 2018; Oliveira et al., 2017).

Therefore, ML supports risk assessment in several ways, integrating heterogeneous data separated in time and space (Zennaro et al., 2021). For what concern pluvial flood risk, different types of ML statistical approaches have been adopted and include Logistic Regression (Ettinger et al. 2016), Frequency Ratio (Tehrany et al., 2015a; Rahmati et al., 2016) and Fuzzy Logic (Pierdicca et al. 2010). Non-linear ML algorithms used in flood susceptibility models include Artificial Neural Network (Kia et al. 2012), Random Forest (Wang et al., 2015; Chapi et al., 2017; Gudiyangada Nachappa et al. 2020a; Gudiyangada Nachappa et al. 2020b), Decision Tree (Tehrany et al., 2015b; Pham et al., 2021; Tehrany et al., 2014; Khosravi et al., 2018; Choubin et al., 2019), Convolutional Neural Network (Ghorbanzadeh et al., 2019a; Wang et al., 2020) and Support Vector Machine (Tehrany et al., 2014; Choubin et al., 2019; Gudiyangada Nachappa et al. 2020a; Gudiyangada Nachappa et al. 2020b; Wang et al., 2020). Any ML model has advantages and disadvantages and the model's performance largely depends on the dataset and its structure (Gudiyangada Nachappa et al. 2020a; Gudiyangada Nachappa et al. 2020b). The prediction of pluvial flood events can be a regression or a classification problem, depending on the purpose of the assessment and on the historical data available. In the preparation phase predictive models are mainly used for hazard mapping and emergency planning and thus information about the spatial extent of flooded areas can be sufficient; in such cases flood prediction is a classification problem, where we predict the occurrence, or not, of a pluvial flood event in a specific space and time. By contrast, during the response and recovery phase maximum flood depth and velocity should be modeled thus requiring a regression analysis. In classification tasks, each point in the area under assessment is characterized by a specific and unique combination of attributes, the so-called triggering factors (e.g., aspect, curvature, distance to rivers, land use, elevation, etc.), which are processed in the ML models. The study area is hence represented by a set of observations characterized by triggering factors information values for each specific location and time point. During an extreme precipitation event, water might not be quickly absorbed by the ground, determining a lengthening of the event that can last more than one day. This fact may lead to temporal correlation between observations at the same location. Furthermore, observations are also spatially correlated, as areas close to each other tend to have the same characteristics, and so, similar probabilities of being affected by pluvial flood events (Meyer et al. 2018). For this reason, standard ML procedures that randomly split the dataset in train and test set could be not suitable for spatio-temporal analysis and classification of pluvial flood risk, as they violate the assumption of independence between train and test data (Hastie et al., 2009), leading to overoptimistic models that underestimate the prediction error. Drawing on such considerations, the overall aim of the analysis presented here is to develop and test a ML-based methodology for rapid pluvial flood risk prediction in the Metropolitan City of Venice (Northern Italy) considering both spatial and temporal constraints. The novelty of the presented methodology is the development of a classification model that reduces the overfitting by introducing a features selection process and by involving spatio-temporal cross-validation. Ke et al., (2020) developed a ML approach to predict the occurrence of pluvial flood in urban areas. They defined a model able to determine the rainfall flooding thresholds, providing a binary result of flooding or not flooding occurrence. Despite the proposed model is built on the 10-fold cross-validation technique according to temporal rainfall accumulation, they did not estimate the spatial component, hence the correlation of the spatial distribution of inundation and inundation depth with the temporal occurrence of the events. Similarly, Guo et al. (2022) developed a flood prediction model capable of generalizing different catchment areas and to generate flood predictions. The data-driven model is built on one rainfall event and considers maximum water depth and flow velocity as key factors, while neglecting the temporal variable. The Convolutional Neural Network model was used for spatializing flood prediction also under different scenarios. However, a drawback of the model is that they performed a holdout validation instead of a k-fold

cross validation, since it would have smoothed out the results and avoided misleading results, which was compensated with a large dataset. Furthermore, the model was oversimplified by focusing only on one main contributing factor (i.e., the elevation) and neglecting others as land cover and spatial distribution of precipitation intensity. The model performed a good prediction accuracy and was able to generalize on different catchment areas and topographies, despite being limited to few affecting features. Overall, multiple studies have developed models able to detect the spatial component with accuracy, even though a spatio-temporal cross validation technique would be more reliable in reproducing and predicting unknown flood events.

Drawing on such considerations, the methodology propose demonstrates how spatio-temporal cross validation should be used to improve the accuracy of pluvial flood mapping both across space and time, a necessary improvement especially to support flood prediction over future scenarios and under a changing climate in which the spatial patterns, intensity and frequency of extreme precipitation events could be extremely different from current ones.

A further novelty is the selection of features (i.e., triggering factors) that might affect the occurrence of pluvial flood in the cross-validation process and the consequent discard of those features that are not directly linked to flooding events.

The final output is represented by a suite of risk maps of the flood-prone areas in the Metropolitan City of Venice. Indeed, in the past, the Metropolitan City of Venice has been hit by recurrent extreme pluvial flood events and, given the future climate scenarios and the urbanization rate, worsening conditions and extreme events are expected, requiring therefore an assessment of pluvial flood risks and the mapping of the most vulnerable areas to identify suitable measures to include in the definition of the emergency and adaptation plan, currently under development.

After a brief introduction of the case study area, the historical pluvial flood extreme events and the available input data (Section 2), the paper describes the cross-validation approaches and ML models applied to the case study area to create the pluvial flood risk mapping (Section 3), and finally discusses the prediction error with different models and the risk mapping for 60 historical extreme events in the Metropolitan City of Venice (Section 4).

2. Case study area

2.1. The metropolitan city of venice

The Metropolitan City of Venice (CMV) is located in Veneto region, in the North East coast of the Adriatic Sea. It is placed in the Venetian Plane, consists of 44 municipalities and has an overall extension of 2467 km² with a population of 848.829 inhabitants (ISTAT, 2020) (Fig. 1b, Figure A Supplementary Material), mainly concentrated in the west side of the urban area, adjacent to the city of Venice. Multiple economic activities take place in the study area, ranging from industry (i.e., Mestre and Marghera district), tourism, seaport activities, shipping and fishing along the coastal area (Dipartimento per gli Affari Regionali e le Autonomie 2017). The CMV is a highly urbanized area, whose development was driven by the economic growth after World War II and the expansion of the petrochemical plant of Porto Marghera. The rapid urbanization rate led, in the past, to a growing demand of residential spaces and infrastructures, determining radical changes of entire areas of the city and causing the rapid conversion of green areas into urban zones. The extended development determined structural hydraulic criticalities (Venice Province, 2011) that, in recent years, the administrators tried to overcome also through environmental restoration actions, such as the creation of green public areas (e.g., Bissuola Park, Mestre Wood, San Giuliano Park). However, despite the implementation of such measures, there are still many criticalities due to the inappropriateness of the sewer and drainage systems, which became evident in the last decade when different portions of the urban area have been

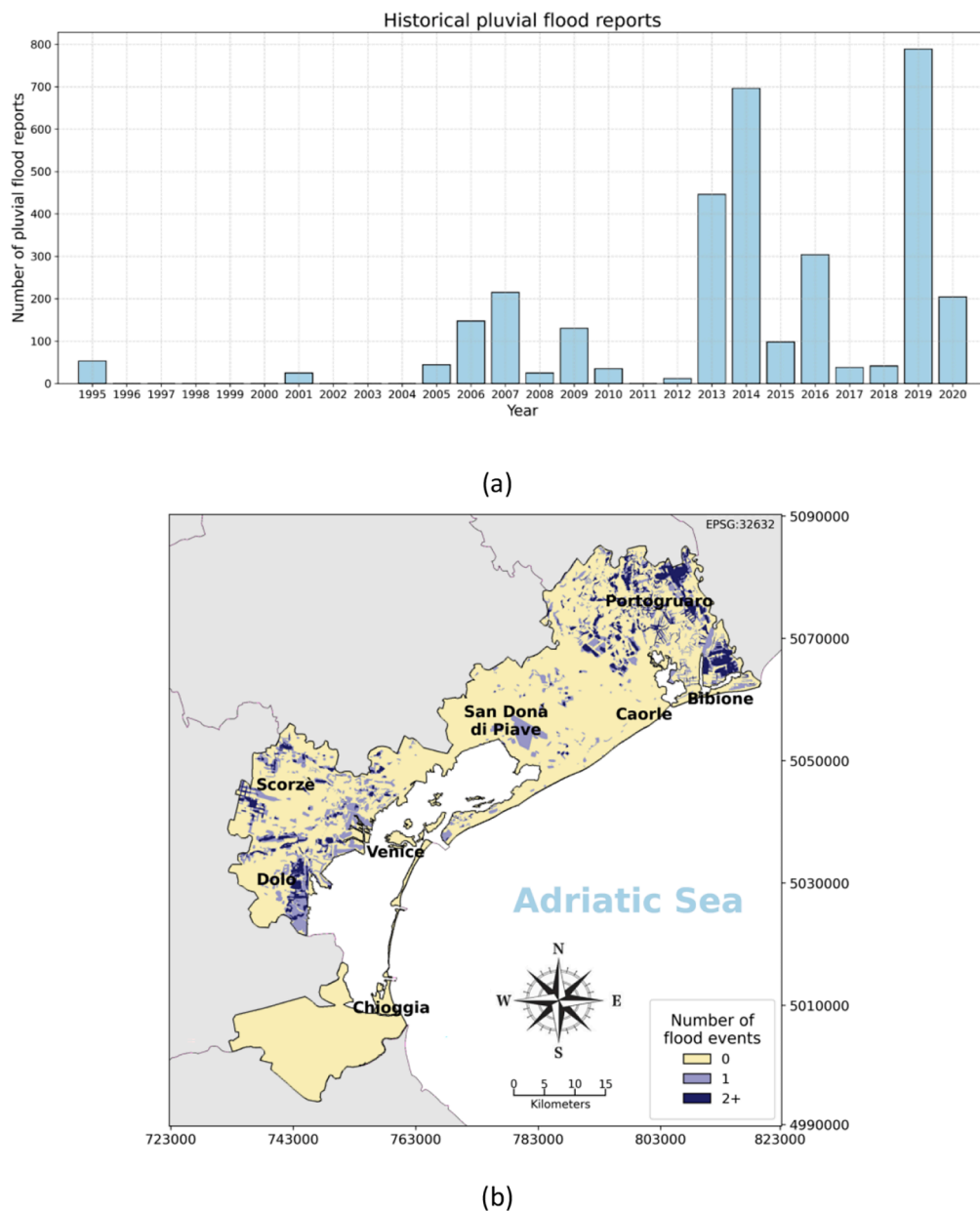


Fig. 1. Number of pluvial flood events per year in the period 1995–2020 (a) and spatial distribution of the total events of flood reported in the Metropolitan city of Venice (b).

flooded in concomitance with severe rainfall events. The increasing frequency and intensity in extreme rainfall events expected for the Mediterranean region (Blöschl et al., 2019) exposes the study area to a potential increase of pluvial flood risk. Specifically, climate projections foreseen for the long term a reduction of precipitation for the summer period and an increase during the winter months in Northern Italy (Spano et al., 2020). According to the National Adaptation Plan to Climate Change (PNACC, 2018), extreme precipitation events are expected to occur for more than 100 days during the winter season as well as more than 20 days per year of precipitation above the 95th percentile in the area of concern. Beside the climatic drivers, the social and structural components (e.g., the urbanization degree, the rate of population growth in urban areas, the appropriateness of the drainage and sewer systems) as well as the environmental, geological and hydrological features of the territory will directly contribute to increase the vulnerability of the territory to pluvial floods events influencing the soil absorption rate and the water runoff (Huong and Pathirana, 2013; Spano et al., 2020).

2.2. Data collection and preprocessing

ML models are data-driven tools that offer an automated means of recognizing patterns in high-dimensional data (Cracknell and Reading, 2014). ML models link a set of independent variables to the response variable and use only the independent variables for making predictions. In predicting pluvial flood events, ML models learn from variables, called *triggering factors*, to predict a *response variable* at a different time and location, within a defined model domain (Meyer et al. 2018). In the pluvial flood risk model developed for the MCV, the response variable is represented by flooded areas, while the triggering factors are represented by territory's features that might positively or negatively affect the flooding occurrence. The triggering factors have a direct impact on the accuracy of the models in predicting pluvial flood areas. Therefore, the collected data (e.g., triggering factors and the pluvial flood areas) constitute the dataset, used to build up the model.

2.2.1. Historical events inventory

The historical events inventory reports the locations of the pluvial flood areas and provides detailed information about the characteristics of historical flood events (Zazo et al., 2018). A dataset of historical events recorded between 1995 and 2020 was provided by the Metropolitan City of Venice (<https://webgis2.cittametropolitana.ve.it>) with detailed information regarding the date, localization, the municipalities affected, the extension of flooded areas and the cause (i.e. heavy rain, coastal or riverine flood) for each event. The dataset has been built collecting the reports made by citizens hit by the flooding event to the firefighters.

Fig. 1a shows the number of historical events that occurred each year from 1995 to 2020. Fig. 1b represents the spatial distribution of flooded areas according to their frequency in the MCV. The number of pluvial flood events has consistently increased in the last decade and a net peak has been registered in 2014 and 2019. Most of the historical events reported in the case study area are close to the cities of Portogruaro and Bibione, in the North, and close to Camponogara and Mestre in the South-Center of the MCV.

To build up the ML model, a sampling of 40 historical events were selected among the most relevant pluvial flood events (i.e., when at least ten pluvial flood areas have been reported to the authorities). In order to be sure to include only pluvial flood events, only flood events associated with heavy rainfall have been selected from the historical dataset while riverine, coastal or compounded flood have been excluded.

Moreover, other 20 non-flood historical events in the time period 1995–2020 were randomly selected and added to the dataset to get a more exhaustive overview of the relationship among flooded areas, non-flooded areas and triggering factors.

2.2.2. Flood triggering factors

Based on literature (e.g., Arabameri et al., 2019; Wang et al., 2020; Shahabi et al., 2021) and available data, 12 triggering factors were initially selected as relevant for the assessment of pluvial flood risk in the MCV. Table 1 summarizes the main characteristics of the triggering factors, including their spatial and temporal resolution, the number and

the type of features derived from each variable, and the data source used for their characterization. In fact, the variables can be classified in numerical, if the values are numbers, or in categorical, if the values present or not a specific class of the variable itself.

Aspect is the direction or orientation of the maximum slope of the surface. Different directions may face different effects in terms of precipitation and solar radiation thus affecting hydrological conditions (Bathrellos et al., 2018; Wang et al., 2020). Four categories of aspect have been considered according to different angle ranges: North (45° – 135°), West (135° – 225°), South (225° – 315°), and East (315° – 45°).

Curvature is the degree of distortion of the slope surface computed from the Digital Terrain Model (DTM) and reflects the morphology of a given area (Mojaddadi et al., 2017; Wang et al., 2020). Curvature has been differentiated into three categories: flat, concave or convex.

Elevation has an inverse relationship with flooding (Fernandez and Lutz, 2010; Shahabi et al., 2021). Lower elevations may lead to higher probability of pluvial flood (Bui et al., 2019). Elevation was treated as a numerical feature.

Slope represents the inclination of the surface and has a direct effect on flooding because it affects surface runoff, its velocity and water infiltration rates (Costache et al., 2020). Accordingly, the lower the slope, the higher the rate of infiltration and the lower the flow velocity. Slope was computed from the Digital Terrain Model (DTM) and was differentiated into two categories: presence or absence of slope.

Soil is the classification of specific areas based on their morphology, granulometry of sediments and age of surfaces (Carta dei Suoli del Veneto, 2019). Different soils imply different permeability and thus, different flood potentials. Soil features have been collected from ARPAV database and were differentiated into four categories: low ancient plain (BA), recent low plain (BR), coastal plain (CL) and urban areas (Urb).

Texture is a soil infiltration property, classified by the percentage composition of its solid particles distinguished in granulometric categories and it is important in determining the total amount of surface runoff during rainfall events (Kaspersen et al., 2017). Texture indicates the soil material at 50 cm under the surface and it was differentiated in 10 categories: clayey, clayey frank, clayey sandy frank, clayey silty, clayey

Table 1

Triggering factors selected, response variable and sources used for their characterization. Digital Terrain Model (DTM) and Agenzia Regionale per la Prevenzione e Protezione Ambientale del Veneto (ARPAV).

Variable name	Spatial resolution	Temporal resolution	Features	Variable Type	Data source
Aspect	25 m	2019	4	Categorical	DTM Veneto Region (https://idt2.regione.veneto.it/idt/downloader/download)
Curvature	25 m	2019	3	Categorical	DTM Veneto Region (https://idt2.regione.veneto.it/idt/downloader/download)
Distance to river	–	2020	4	Categorical / Numerical	Metropolitan City of Venice (https://webgis2.cittametropolitana.ve.it)
Distance to road	–	2019	4	Categorical / Numerical	Metropolitan City of Venice (https://webgis2.cittametropolitana.ve.it)
Distance to sea	–	2016	4	Categorical / Numerical	ARPAV (https://geomap.arpa.veneto.it/layers)
Elevation	25 m	2016	1	Numerical	DTM Veneto Region (https://idt2.regione.veneto.it/idt/downloader/download)
Flood events inventory	–	1995–2020	60	Categorical	Metropolitan City of Venice (https://webgis2.cittametropolitana.ve.it)
Land use	100 m	1990–2018	25	Categorical	CORINE Land Cover (https://land.copernicus.eu/pan-europea/corine-land-cover)
Normalized Difference Vegetation Index (NDVI)	10 m	2015	1	Numerical	Sentinel 2 - Earth Explorer (https://earthexplorer.usgs.gov/)
Permeability	–	2016	5	Categorical	ARPAV (https://geomap.arpa.veneto.it/layers)
Precipitations (daily cumulative)	188 stations	1995–2020	60	Numerical	ARPAV (private data)
Precipitations (hourly maximum intensity)	188 stations	1995–2020	60	Numerical	ARPAV (private data)
Slope	25 m	2019	1	Categorical	DTM Veneto Region (https://idt2.regione.veneto.it/idt/downloader/download)
Soil	–	2015	5	Categorical	ARPAV (https://geomap.arpa.veneto.it/layers)
Texture	–	2016	11	Categorical	ARPAV (https://geomap.arpa.veneto.it/layers)

silty frank, frank, organic materials, sandy, sandy frank, sandy loam and silty frank.

Permeability measures the capacity of a surface to absorb or reject water, thus it is an indicator of water infiltration. Permeability has been differentiated in 5 categories reporting the permeability range values, expressed in millimeters per hour (mm/h): very low (0,036–0,36 mm/h), low (0,36–3,6 mm/h), medium (3,6–36 mm/h), high (36–360 mm/h), very high (360 + mm/h).

Land use can affect water absorption and thus runoff (Santos and Reis, 2018). Land use data have been extracted from the CORINE Land Cover dataset for five years: 1990, 2000, 2006, 2012, and 2018. The 44 land types were aggregated into five main categories: agriculture areas, green areas, industrial areas, urban areas and water areas.

Normalized Difference Vegetation Index (NDVI) is used to study the vegetation density of the land surface (Rouse et al. 1974). Changes in NDVI indicate modifications in vegetation (Ahmed and Akter, 2017); it can be an important triggering factor as higher vegetation densities are assumed to have lower probabilities of flooding within the study area (Chapi et al., 2017). NDVI data were extracted from three Satellite Sentinel2 images (i.e., tiles T32TQQ, T32TQR and T33TUL) taken on May 5, 2015 and downloaded from EarthExplorer website. NDVI is calculated as:

$$NDVI = (B08 - B04) / (B08 + B04)$$

where B08 is the Near Infrared Band and B04 is the Red Band. NDVI can vary between -1 (low vegetation density) and +1 (high vegetation density).

Precipitation is the main triggering factor contributing to pluvial flood risk. Precipitation data were provided by the Regional Agency for Environmental Prevention and Protection of Veneto (ARPAV) and the cumulative daily precipitation and hourly maximum intensity was extracted for each of the 60 historical events. Since the meteorological stations do not cover all the study area, the spatial distribution was computed by interpolating the 188 stations placed across Veneto region using Linear Ordinary Kriging method (Li et al., 2011; Yang, 2015). Precipitation data were treated as numerical features.

Distance to road is an important feature to consider as the proximity to artificial infrastructure can impose a negative impact on pluvial flood vulnerability. Indeed, roads can represent a proxy of highly impervious areas leading to higher probability of pluvial flood (Gaitan et al., 2016). Distance to road has been differentiated in three categories (0 m – 250 m, 250 m – 500 m, greater than 500 m) and it was also treated as a numerical feature.

Distance to river indicates the Euclidean distance in meters from river networks. Areas close to rivers are generally considered more susceptible to flooding compared to more distant areas (Chapi et al., 2017) as rivers represent main pathways for flood discharge and expansion. Distance to the river has been differentiated in three categories (0 m – 250 m, 250 m – 500 m, greater than 500 m) and it was also treated as a numerical feature.

Distance to sea was assessed because the case study includes areas close to the coast. The distance to sea has been differentiated in three categories (0 m – 250 m, 250 m – 500 m, greater than 500 m) and it was also treated as a numerical feature.

Each variable described above has a different unit of measure. These differences between features require rescaling the values in order to ensure that all inputs are equally comparable when developing the ML models (Hastie et al., 2009). Therefore, all categorical features were treated as binary features with a value of 1 if the category is present in that observation, or 0 otherwise, while the numerical features were scaled between 0 and 1, except for NDVI that is included between -1 and +1. The spatial distribution of the triggering factors in the case study domain is represented in Figure B (Supplementary Material).

3. Methods

3.1. Machine learning models

The pluvial flood risk assessment methodology proposed here allows a first assessment of the areas that are more likely to be affected by pluvial flood in the Metropolitan City of Venice given specific precipitation events and the characteristics of the territory. Pluvial flood prediction, in our case study, is a classification task, which aims to predict if an area is likely to be flooded or not, and to explore the relationship existing between the features included in the dataset and the response variable, that does not include the pluvial flood depth. Since historical events of pluvial flood areas are available, supervised ML models can be used to learn the relationship between pluvial flood areas (i.e., the response variable) and the triggering factors (i.e., the features) to identify the most relevant patterns in the data. In this paper, three different ML models were considered and compared, namely Logistic Regression, Neural Networks and Random Forest.

3.1.1. Logistic Regression

Logistic Regression (LR) is a widely used model for classification (i.e. when the response variable is dichotomous) (Azzalini and Scarpa, 2012). LR is expressed as a linear combination $\eta(x)$ of the selected features:

$$\eta(x) = \beta_1 x_1 + \dots + \beta_p x_p = x^T \beta \quad (1)$$

where x_i , $i \in \{1, \dots, p\}$ are the p features and β_i , $i \in \{1, \dots, p\}$ are the p weights. From $\eta(x)$, it is possible to calculate the probability π of pluvial flood occurrence as:

$$\pi = \frac{e^{\eta(x)}}{1 + e^{\eta(x)}} \quad (2)$$

where $0 \leq \pi \leq 1$.

3.1.2. Neural Networks

A Neural Network (NN) is a complex ML model that approximates almost any nonlinear function (Azzalini and Scarpa, 2012). Its basic architecture is made up of three layers: i) an input layer that includes the features; ii) one hidden layer that links the input layer with the output layer using nonlinear functions and optimized weights; iii) an output layer that includes the response variable. A NN model is particularly used for its flexibility and good performances, however it does not allow to identify the most important features that affect the model, leading to a weak interpretation of the results.

3.1.3. Random Forest

Random Forest (RF) is an ensemble classification model that averages the predictions of a number k of classification trees (Breiman, 2001). RF combines different models to decrease the unstable results arising from a single classification tree prediction, which might be affected by a specific subset of data selected for the training set (Hastie et al., 2009). Indeed, each classification tree is fit using a random subset of features and observations, which make the final ensemble model more reliable.

3.2. ML-Based methodology for pluvial flood risk mapping

As in the assessment of pluvial flood risk the triggering factors and the response variable vary both in time and in space, a specific methodology considering spatio-temporal constraints needs to be developed.

The flowchart in Fig. 2 is meant to guide the reader through the methodology applied to identify pluvial flood risk areas.

The aim of the proposed approach is to develop a ML model that is reliable to predict pluvial flood occurrence in different locations and time periods. For this reason, the triggering factors and the historical

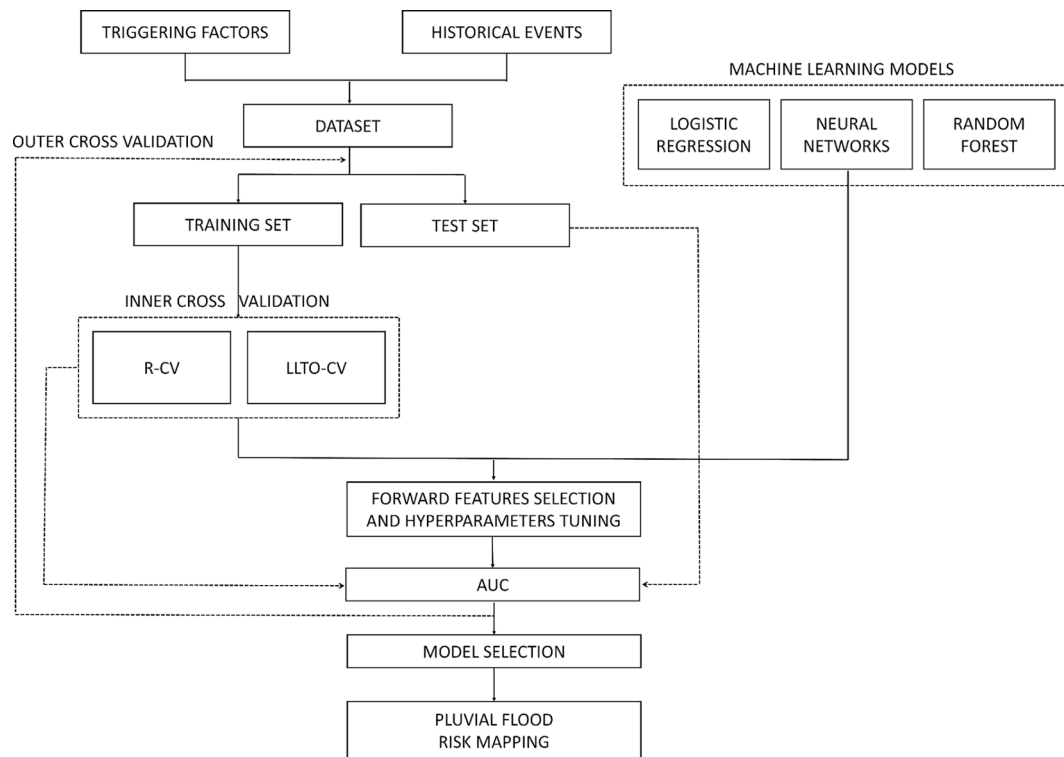


Fig. 2. The flowchart of the ML based methodology to predict pluvial flood risk.

events introduced in Section 2 were collected for the case study to create the dataset used to fit the ML models. The dataset, in ML approaches, is usually split in two folds: training and test set. The training set is used to fit the parameters of the model, while the test set provides an unbiased evaluation of the prediction errors of the model fit on the training set. However, a frequent problem in ML models is overfitting (i.e. when the model has a high performance in predicting training data, but it fails to predict new data that the model has never seen before) (Hastie et al. 2009). A way to reduce overfitting is to split the training set in two parts: the training and the validation set. The validation set is typically used to estimate the prediction error on independent data and to optimize the hyperparameters of the ML models. The hyperparameters of a ML model define its structure and are used to control the learning process of the model. This process is called hyperparameters tuning, that is the research of the ideal model architecture (Lecun et al., 2015).

In this specific case study, the validation set is used for hyperparameters tuning but also to select the most important features to consider to improve the pluvial flood area prediction.

A limitation of splitting the dataset in training, validation and test set might occur when working with spatial data because the assumption of independence between these three folds is not respected in a geographical domain (Roberts et al., 2017). Furthermore, when the dataset is scarce, cross-validation is applied to measure the model's accuracy to predict new data in more than one configuration of training, validation and test set. In this study, since the data of flooded and non-flooded areas in the CMV were limited, cross-validation was applied to provide a better appraisal of the prediction error. Cross-validation can be distinguished in outer and inner cross-validation. Outer cross-validation allows different configuration of test set, while inner cross-validation allows different configuration of training and validation set. Moreover, the inner cross-validation methods, in this case study, are distinguished in random cross-validation (R-CV) and spatio-temporal cross-validation (LLTO-CV) to compare the first approach, widely used in literature, with a different approach that takes into consideration both space and time to evaluate the performances of the model.

As described in Fig. 2, triggering factors and historical events data

were preprocessed to construct the dataset, which has been split in training and test set. Then, two different inner cross-validation approaches, R-CV and LLTO-CV, were used to additionally split the training set into training and validation set. Three different machine learning models were thus applied, using both cross-validation techniques, in order to select the best features and hyperparameters based on the Area Under the Curve (AUC) score in the validation set. AUC is a measure of accuracy commonly used to evaluate machine learning models in binary classification problems and it ranges between 0 and 1, where 1 indicates the maximum accuracy score. In addition, an outer cross-validation was applied leading to the reiteration of the whole process until all the observations in the dataset have been part of the test set exactly once. Finally, the best model was selected and used to predict pluvial flood events based on the best AUC score in the validation set.

3.2.1. Dataset folds

ML models were compared based on their accuracy to spatial-temporally predict a new pluvial flood event. Hence, the dataset was split in $T \times S$ folds, where T is the number of time periods in which the historical events are grouped and S is the number of spatial areas in which the case study areas are split. Time intervals and spatial areas were chosen to guarantee an almost equal distribution of pluvial flood areas in each fold. Indeed, the time periods were calculated using the quantile distribution of the flooded observation dates. In particular, the limits of the time period classes are the $0/T$, $1/T$, ..., T/T quantile of the historical events with pluvial flood areas reported, where the $0/T$ quantile is the first event and the T/T quantile is the last one. Regarding the spatial areas, pluvial flood areas were labeled according to the location (i.e. the city where the flood occurred), while the latitude of the city center was used as criteria to split the areas in S groups. In particular, the quantile distribution of the latitude of cities in which the pluvial flood occurred was used to split the data in S spatial areas, and the limits of the spatial classes were the $0/S$, $1/S$, ..., S/S quantile of the cities latitude, where the $0/S$ quantile is the highest latitude and S/S quantile is the lowest one. Once the folds were defined, the dataset was split into training and test sets.

3.2.2. Cross-Validation

To get a reliable estimation of the model performance that is not biased by a specific configuration of training and test set, we applied a nested cross-validation. Nested cross-validation is an approach for hyperparameter optimization, features selection and model selection that attempts to overcome the problem of overfitting the training set. Hence, cross-validation is applied in order to select the most important variables, to optimize the hyperparameters of the models and to select the most appropriate model. Nested cross-validation includes two typologies of cross-validation: the outer cross-validation used to estimate the prediction error in different configurations of the training and test set, and the inner cross-validation used for hyperparameters tuning and to select the most important features based on the model performance in different configuration of training and validation set.

3.2.2.1. Outer leave location and time-out cross-validation. Outer cross-validation allows to take advantage of all the available data to measure the accuracy of the models in different test sets. As the three ML models are compared by their ability to predict a new spatio-temporal event, outer leave location and time-out cross-validation was considered. At each step, one fold k^{ts} between the $T \times S$ folds is selected to be part of the test set, $(T-1) + (S-1)$ folds that have in common the time period or the spatial area with the test set were excluded from the analysis while the $(T-1) \times (S-1)$ independent folds are selected to be part of the training set, until all folds have been exactly once part of the test set. Then, the training set used to fit the model was split into a training and validation set to apply forward features selection and hyperparameters tuning. Fig. 3a shows the outer cross-validation procedure in the case of $T = 4$ and $S = 4$. At each step, one of the 16 folds was used as test set, while 9 independent folds were part of the training set. The same number of folds were used in the case study, which means that at each step roughly 56.25% of the data were used as training set and 6.25% as test set for outer cross-validation, while the remaining 37.5% of the data were excluded.

3.2.2.2. Inner Cross-Validation. The inner cross-validation is applied to select the most relevant features and to optimize the hyperparameters based on the performances of the model in different configurations of training and validation set. In this paper, inner cross-validation is differentiated in Inner Random cross-validation and Inner Leave Location and Time-Out cross-validation. We applied both methodologies to understand which one better performs in forecasting new flooded areas in consequence of extreme events.

In the Inner Random cross-validation (R-CV) approach, the training set is randomly split in k_R roughly equal-size folds. At each step, one of the k_R folds was used as validation set and the remaining $k_R - 1$ folds were used to fit the model, until all the folds have been part of the validation set exactly once. A limitation of R-CV is the assumption of independence between training and validation observations, that, if it is not satisfied, leads to optimistic accuracy in the validation set where the model selection will favor the most complex models (Roberts et al., 2017). In pluvial flood problems, the assumption of independence could not be satisfied since observations are correlated both in time and space, hence they cannot be split in training and validation set randomly. This approach determines that R-CV could overestimate the predictions accuracy in the validation set, as it is correlated with the training set, but it may lead to bad predictions in an independent test set from a different spatio-temporal domain that the model has never seen before. Fig. 3b shows the inner random cross-validation procedure in the case of $T = 4$ and $S = 4$, where the test set is set to be the $k^{1:1}$. The 9 folds in the training set, independent from the test set, are merged and split in 4 random folds. When cross-validation is applied, at each step, one of the random folds is chosen as validation set and the remaining 3 folds are used as training set to fit the model until all the folds have been part of the validation set exactly once.

Inner Leave Location and Time Out cross-validation (LLTO-CV) is an approach that faces the spatio-temporal problems considering both time and space in the creation of the folds for the training and validation set. Indeed, the training set initially includes $(T-1) \times (S-1)$ folds independent from the test set and, when cross-validation is applied, at each step, a fold k^{ts} folds is chosen to be part of the validation set, $(T-2) + (S-2)$ folds that have in common the time period or the spatial area with the validation set were excluded from the analysis and the remaining $(T-2)(S-2)$ folds are used to fit the ML model. The procedure is repeated until all the folds have been part of the validation set exactly once. Fig. 3c shows a LLTO-CV example for $T = 4$, $S = 4$, and where $k^{1:1}$ fold is chosen as test set. At each step, one of the 9 folds of the training set is chosen as a validation set, 4 folds are discarded and 4 independent folds are used to fit the model. The same number of folds were used in the case study, which means that at each step roughly 44.44% of the data were used as training set and 11.11% as test set for inner cross-validation, while the remaining 44.45% of the data were excluded.

3.3. Forward features selection and hyperparameters tuning

Usually cross-validation is used to estimate the model performances in predicting different validation sets, but also to research the best architecture of the model. The research of the architecture that maximizes the accuracy of the model in the validation set is called hyperparameters tuning, as one or more parameters of the model are optimized to find the best configuration. In this paper, we add to the hyperparameters tuning the research of the best features that improves the accuracy of the model in predicting pluvial flood areas. In this procedure, called forward features selection, at each step, the best feature is added to the model until no more features are useful to improve predictions.

3.3.1. Forward features selection

Most of the features collected in the dataset describe the properties of a location (e.g., curvature, elevation, slope, etc.). These features vary in space but not in time, which means that each pixel has a unique combination of static features and it has unique properties, comparable with quasi-unique IDs of the pixels (Meyer et al., 2018). Features that lead to overfitting can be automatically identified and removed using a forward feature selection method to exclude the misleading ones from the models (Meyer et al., 2018). The forward features selection method, at each step, includes in the model the feature that mostly improves the accuracy of the model, in classification tasks, or mostly reduces the prediction error, in regression tasks, until no improvements are shown when adding a new feature. In this case study, at each step, the feature that mostly improves the AUC, a recommended metric used instead of the overall accuracy to compare different ML algorithms in classification problems (Bradley and Andrew, 1997), was introduced. The forward features selection process ends when no more features can add enough value to the model to improve the AUC score at least by 0.01 with respect to the previous step. This methodology strongly reduces the number of features by selecting the most representative ones, resulting in simpler and more understandable models.

3.3.2. Hyperparameters tuning

Hyperparameters setting is a fundamental step in building machine learning models (Lecun et al., 2015). Some ML models need to optimize specific hyperparameters to find the best architecture that maximizes the AUC score through inner cross-validation. As the forward features selection methodology leads to a sensible increase of the computational time to fit the models, hyperparameters are fixed before the features selection process and the tuning is computed only at the end of the process, even though the procedure does not guarantee to obtain the best hyperparameters configuration. Indeed, for each of the ML models initial hyperparameters are manually set. In particular, Logistic Regression is set with a regularization hyperparameter equal to 1, Random Forest nodes are expanded until all leaves are pure and Neural

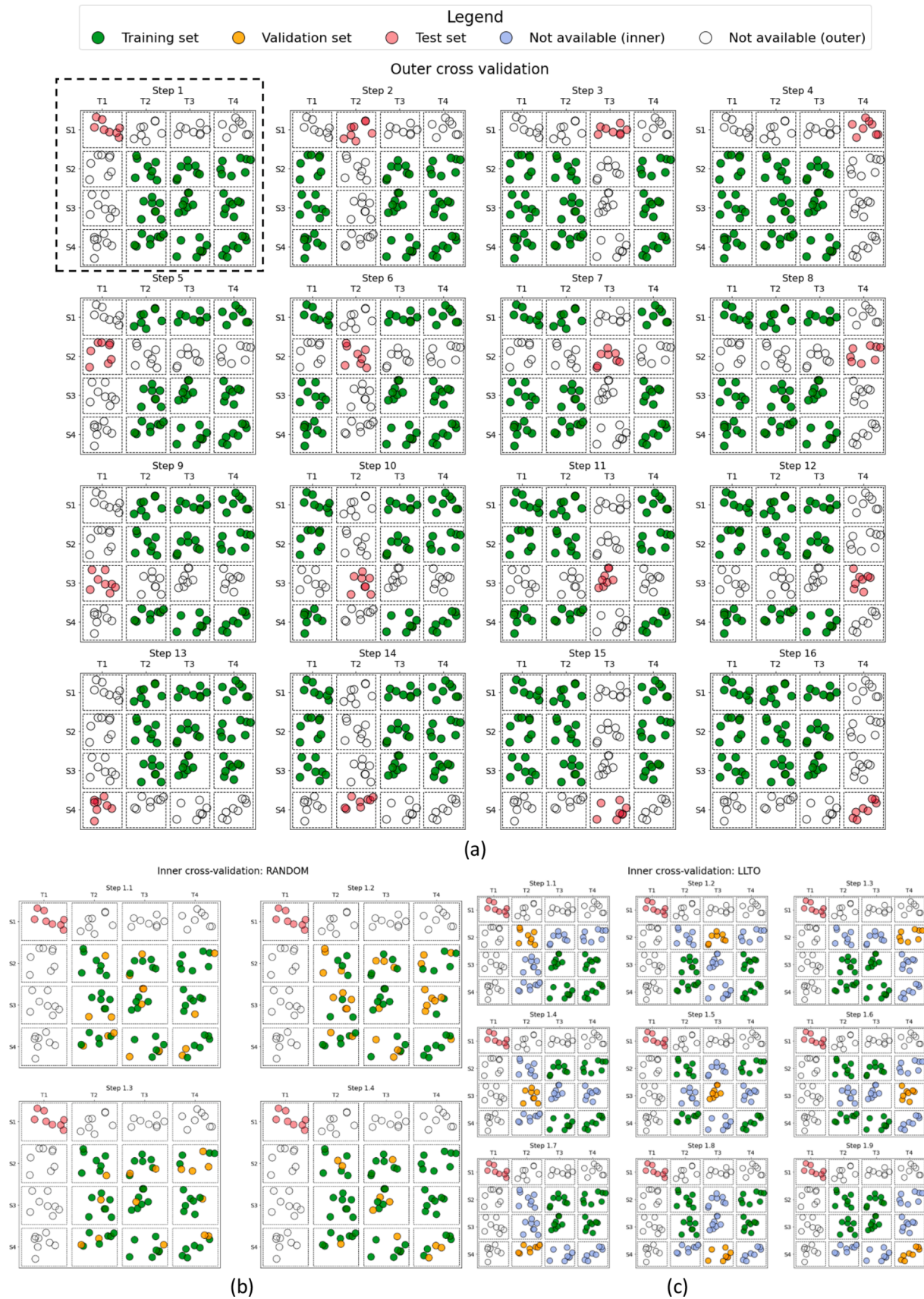


Fig. 3. (a) Outer cross-validation; (b) Inner random cross-validation; (c) Inner leave location and time out cross-validation.

Networks are set with one hidden layer and 2 nodes. At the end of the forward features selection process, hyperparameters tuning is computed to research the best configuration of each model, considering:

- a regularization hyperparameter $C = 0.01, 0.1, 1, 10$ or 100 for Logistic Regression;
- one hidden layer with a number between 2 and 9 nodes for Neural Networks;
- a maximum depth between 2 and 20 nodes for Random Forest.

The hyperparameters that are not considered in the hyperparameters tuning process are set to be equal to their standard value defined by the package *sklearn* in Python.

3.4. Model selection

The three ML models introduced in Section 3.1 were fitted with two different cross-validation approaches and, following the flowchart in Fig. 3, were compared for the accuracy assessment through the AUC score. The AUC is a measure of accuracy of the predictions compared to the real values and it is computed using the validation set. As all the observations of the training set become part of the validation set exactly once, the AUC can evaluate the model accuracy including all the observations available in the training set. Hence, at the end of the inner cross-validation, the best features and hyperparameters are chosen based on the AUC score. But if the model selection is based only on one AUC score, the selected model is biased by a specific configuration of the training set and test set. For this reason, outer cross-validation returns $T \times S$ different configurations of the training and test set, that permit the computation of $T \times S$ AUC score. Then, the model with the highest AUC score averaged between the $T \times S$ folds is selected to perform pluvial flood risk mapping.

3.5. Pluvial flood risk mapping

The ML model selected in the previous step was finally tested to check its ability in estimating the occurrence of pluvial flood in selected historical events and thus to create a pluvial flood risk map. Each map represents the probability that a pluvial flood occurs in a specific pixel of the study area. The final aim was to investigate if the model is able to properly reproduce the events occurred in the case study and thus, if it is suitable to be used for future predictions.

4. Results

4.1. Dataset and Cross-Validation

The dataset was created from 60 historical events that includes 47 219 280 observations (i.e., pixels that represent the case study area in different days). The number of pluvial flood areas is much lower than the non-flooded areas, as only 0.49% of the observations represent pluvial flood areas (231'420 pixels). In this case, the dataset is unbalanced so, in order to reduce the number of non-flooded areas, a random subset of 10'000 non-flooded observations for each of the 60 dates in addition to all the flooded observations was selected. Accordingly, the final dataset includes 831'420 observations (600'000 non-flooded pixels and 231'420 flooded pixels) and 47 attributes (46 features and the response variable).

The dataset was split in four time periods (T) and in four spatial locations (S), for a total of 16 folds ($S \times T$). The T folds were created by dividing the events with flooded observations in quartiles and grouped as follows:

- (1995–01-01; 2007–09-30]: it includes 28% of the flooded pixels;
- (2007–09-30; 2014–01-31]: it includes 21% of the flooded pixels;
- (2014–01-31, 2016–02-29]: it includes 23% of the flooded pixels;

- (2016–02-29, 2020–12-31]: it includes 28% of the flooded pixels.

For the spatial distribution, the S groups were created according to the latitude of the city's center. The spatial folds, are the following:

- S1: it includes 27% of the flooded pixels;
- S2: it includes 24% of the flooded pixels;
- S3: it includes 28% of the flooded pixels;
- S4: it includes 21% of the flooded pixels.

In summary, there are $T = 4$ time periods folds and $S = 4$ spatial folds, resulting in $T \times S = 16$ spatio-temporal folds (Figure C and Table A, [Supplementary Material](#)).

Following the methodology described in Section 3.2, one of the 16 folds was chosen as the test set, 6 folds were excluded as they have time or space in common with the test set, and the remaining 9 folds were part of the training set.

The outer cross-validation allowed the models to fit on 16 different configurations of training and test set, as at each step, a different fold was part of the test set. Then, when inner random cross-validation was applied, the 9 folds of the training set were merged together, the observations were shuffled, and 4 roughly equal flooded pixels size folds were created. On the other side, when inner spatio-temporal cross-validation was applied, at each step, one of the 9 folds was used as validation set while the remaining $(T-2) \times (S-2) = 4$ folds, that did not have location and time period in common with the test and validation set, were used as training set.

4.2. Forward features selection and hyperparameters tuning

4.2.1. Forward features selection

When applying the forward features selection method, the features selected can vary according to the machine learning model and the observations in the training set. Thereby, as 16 folds exist, also 16 configurations of training and test set are possible, so that each feature can be selected from 0 to 16 times, indicating a proxy of its importance to predict pluvial flood. Fig. 4 shows the most selected features in the two cross-validation approaches for each of the three ML models previously introduced. On the left, the boxplots represent the number of candidate features in each step of 16 feature selection processes for each model and approach. On the right, the dot plots show the features importance based on the number of times they were selected in 16 features selection processes.

In the Logistic Regression model, the forward features selection process selected daily cumulative precipitation for 16 times, making it a key feature for pluvial flood predictions. Another feature considered important by the LR model was the distance to the river, selected 12 times in the LLTO-CV and 13 times in the R-CV. The remaining features were selected less than 50% times, both with the LLTO-CV and R-CV approaches, hence were considered negligible for the model for predicting pluvial flood.

On the other hand, the Neural Network model selected only the daily cumulative precipitation as a key feature, while the remaining ones received low importance scores. Among the remaining candidates, only the hourly maximum precipitation was selected more than 2 times.

Finally, in the Random Forest model with R-CV approach, daily cumulative precipitation, hourly maximum precipitation, distance to sea and elevation were selected as relevant features. Interestingly, neither distance to sea nor elevation resulted relevant in the LLTO-CV approach, so it is possible that these features perform better in a random cross-validation approach, while they fail to predict new spatio-temporal conditions. Hence, in the LLTO-CV approach, the Random Forest model considers the daily cumulative precipitation (10 times), distance to river (6 times), hourly maximum precipitation (6 times) and a sandy texture (5 times) as significant characteristics in estimating pluvial flood.

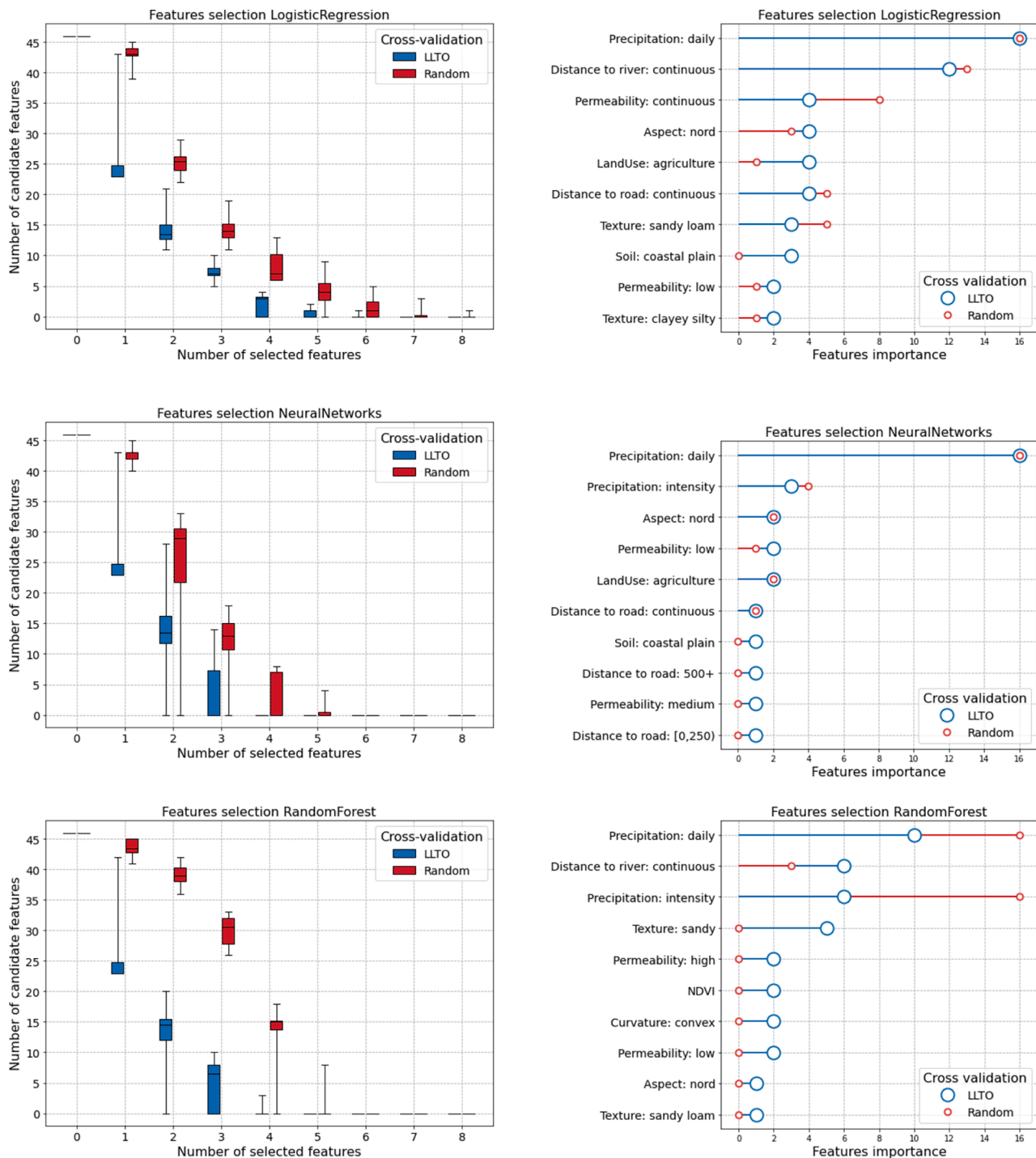


Fig. 4. Forward features selection results.

Overall, the three ML models selected as key features almost the same variables (i.e. daily cumulative precipitation, precipitation intensity and distance to river), simplifying the models' structure and resulting in only three variables useful to predict pluvial flood and affecting the occurrence of flooding events.

Other relevant considerations can be drawn from Fig. 4. At step 0, all the 46 features are suitable to be added to the model. In the next steps, the most important feature is selected, the features that decrease AUC are removed while the remaining features are candidates for the next selection step. It can be noticed that LLTO-CV procedure, for all the three models, has an earlier drop out of the useless features compared to the R-CV approach. R-CV approach tends to build more complex models

keeping more features as candidates in the procedure, until it reaches an excellent AUC score. However, this process leads to overfitting.

4.2.2. Hyperparameters tuning

Hyperparameters tuning output shows clear differences between the two approaches of cross-validation. For Logistic Regression, the median regularization coefficient is 0.1 for LLTO-CV and 1 for R-CV, i.e., a stronger regularization of the coefficient is applied in the LLTO-CV approach. While, for Neural Networks, the median number of nodes of the hidden layer is 6.5 for LLTO-CV and 9 for R-CV. This confirms that R-CV tends to build a more complex Neural Networks architecture using the maximum number of nodes in the available domain imposed on this

hyperparameter. Finally, for Random Forest, the median maximum depth of the tree is 4 for LLTO-CV and 20 for R-CV. The LLTO approach prunes the classification trees to an earlier stage resulting in a simpler model with regards to R-CV. It is worth notice that in the R-CV approach each classification tree is pruned at a depth of 20, which is the maximum value available in the domain imposed to this hyperparameter.

4.3. Model selection

Table 2 reports the AUC score obtained for the training, validation and test set for the two configurations (i.e. R-CV and LLTO-CV) of the three ML models. Differences in the performance of the inner cross-validation approaches can be observed when assessing validation and test set. In the validation set, inner R-CV outperforms inner LLTO-CV for each of the three models, but in the test set, we get the opposite performance. Indeed, in the test set, inner LLTO-CV approach performs better than inner R-CV for each of the three ML models with an AUC score of 0.860 and 0.855 in the LR model, 0.843 and 0.837 in the NN model, and 0.846 and 0.722 in the RF model. These results show the limitation of R-CV in performing spatio-temporal analysis, as it gets excellent performances on the validation set but it fails to predict with accuracy the observations in the test set.

The selection of the most suitable model to predict pluvial flood risk in the MCV was based on the AUC score in the validation set using the LLTO-CV approach, as it is clear that it better approximates the AUC score of the observations in the test set. Hence, the LR model resulted in the best model having the highest AUC score in the validation set and, accordingly, it was selected as the reference model for pluvial flood risk mapping.

The Logistic Regression model was finally implemented for the last features selection by applying an LLTO cross-validation approach to the whole dataset. The final AUC score is 0.864, which means that the model is good at distinguishing between the flood and non-flooded areas across all possible classification thresholds. The best regularization coefficient is 0.05 and, among the 13 triggering factors, the features selected by the model are the daily cumulative precipitation, the distance to river and the distance to roads over 500 m. These three features were tested to detect a possible multicollinearity between them, but the Variance Inflation Factor (VIF) calculated did not find any correlation (all VIF scores lower than 1.07). This model does not consider the remaining 10 triggering factors as relevant to predict pluvial flood events.

The final Logistic Regression model used has the following linear combination $\eta(x)$:

$$\begin{aligned}\eta(x) = & +0.048 \times \text{Precipitation: daily (mm).} \\ & - 0.001 \times \text{Distance to river: numerical (m).} \\ & - 0.560 \times \text{Distance to road: 500+ (m).}\end{aligned}$$

Regarding the feature's importance, the main factor that contributes to the model performances is daily precipitation (AUC = 0.859), while distance to river and distance to road add just 0.005 to the final AUC score. Previous similar ML analysis identified rainfall as the most prominent factor involved in pluvial flooding (Samanta et al., 2018; Meliho, 2022). For instance, Samanta et al. (2018) concluded that rainfall, together with TWI, was the most important flood conditioning

factors in a study that used the frequency ratio technique in the Subarnarekha River basin, India.

Meliho et al. 2022 applied RF, extreme gradient boost (XGB), ANN and KNN to predict vulnerability to flooding in the Ourika watershed in the High Atlas of Morocco. A process of forward feature selection was performed to select variables to be included in each model. Drainage density, rainfall and distance resulted the prominent predisposing factors in the Ourika watershed for most of the considered models.

At the same time distance to river is considered to be one of the most important conditioning factors in most of previous flood susceptibility assessments (Youssef et al., 2016; Ahmadiou et al., 2019; Tien Bui et al., 2019b). Pham et al. (2020) used Credal Decision Tree (CDT) as a base classifier along with four ensemble models for flood modeling in Markazi Province, Iran. They results concluded that distance to river was the most important factor for flood incidence in their study area.

Distance to road has been previously considered only in few studies (Gudiyangada Nachappa et al. 2020a; Gudiyangada Nachappa et al. 2020b). However, Janizadeh et al. (2021) mapped the spatial and temporal variability of flood hazard affected by climate and land-use changes in the future using several machine learning techniques and their ensembles in Iran. According to their analysis, distance to road resulted one the most important feature to predict future flood risk based on the ensemble model.

4.4. Pluvial flood risk mapping

The last step of the analysis consists in creating pluvial flood risk maps for each of the 60 major historical events that occurred in the case study area. The final aim was to investigate if the model is able to properly reproduce events that occurred in the case study and thus, if it is suitable to be used for further predictions (i.e., future risk scenarios).

Therefore, the selected LR model calculated, for each pixel of the study area, the probability of occurrence of pluvial flood, considering the three previously selected features (i.e. daily cumulative precipitation, distance to river and distance to road over 500 m). For a better communication of the results, pixels have been aggregated in six classes of pluvial flood risk: very low (0–20%), low (20–40%), medium (40–60%), high (60–80%), very high (80–95%) and very very high (95–100%). Also, the historical events were aggregated in five classes of percentage of pluvial flood areas: very small (0–1%), small (1–2%), medium (2–3%), high (3–4%), very high (4–5%). Fig. 5 shows the percentage distribution of the pluvial flood risk predicted by the model for the 60 events, as well as the percentage of flooded areas in gray chrome.

The model presents high accuracy and most of the events have been properly predicted, while only few mismatches between the real date of the event and the reported flooding resulted.

Indeed, according to the results reported in Fig. 5, the event that occurred on September 26th 2007 reported the highest percentage of flooded areas (4.24%) and it is classified from the ML model as the 4th biggest event, in terms of number of pixels in the very high-risk class. A good match between the model and the historical event was registered also on the 16th September 2009, whose assessment of the risk corresponded to the historical data.

On the other hand, some events present a mismatch between the model's prediction and the historical data. For example, on September 14th 2006 the model predicted a low risk of pluvial flood occurrence for the whole study area, while the historical data recorded 89 pluvial flood areas for that day. In this example, it is realistic to suppose that the historical data do not correspond to the exact date for the pluvial flood event, that it may be plausible to represent one of the following days. Indeed, the daily average precipitation for September 14th was only 3.3 mm, while the following days it was 50.5 mm on September 15th, 23.6 mm on September 16th and 51.5 mm on September 17th (also represented in Fig. 5).

Similarly, on November 12th, 2014, a gap between the high risk of flooding assessed by the model and the low number of pluvial flood

Table 2
Models performances on train, validation and test set.

Model	Features selected	Cross-validation	Train AUC	Validation AUC	Test AUC
Logistic Regression	4.3	LLTO	0.886	0.876	0.860
Logistic Regression	5.9	Random	0.892	0.892	0.855
Neural Network	2.3	LLTO	0.891	0.833	0.843
Neural Network	3.2	Random	0.886	0.886	0.837
Random Forest	2.6	LLTO	0.908	0.836	0.844
Random Forest	4.1	Random	0.999	0.996	0.722

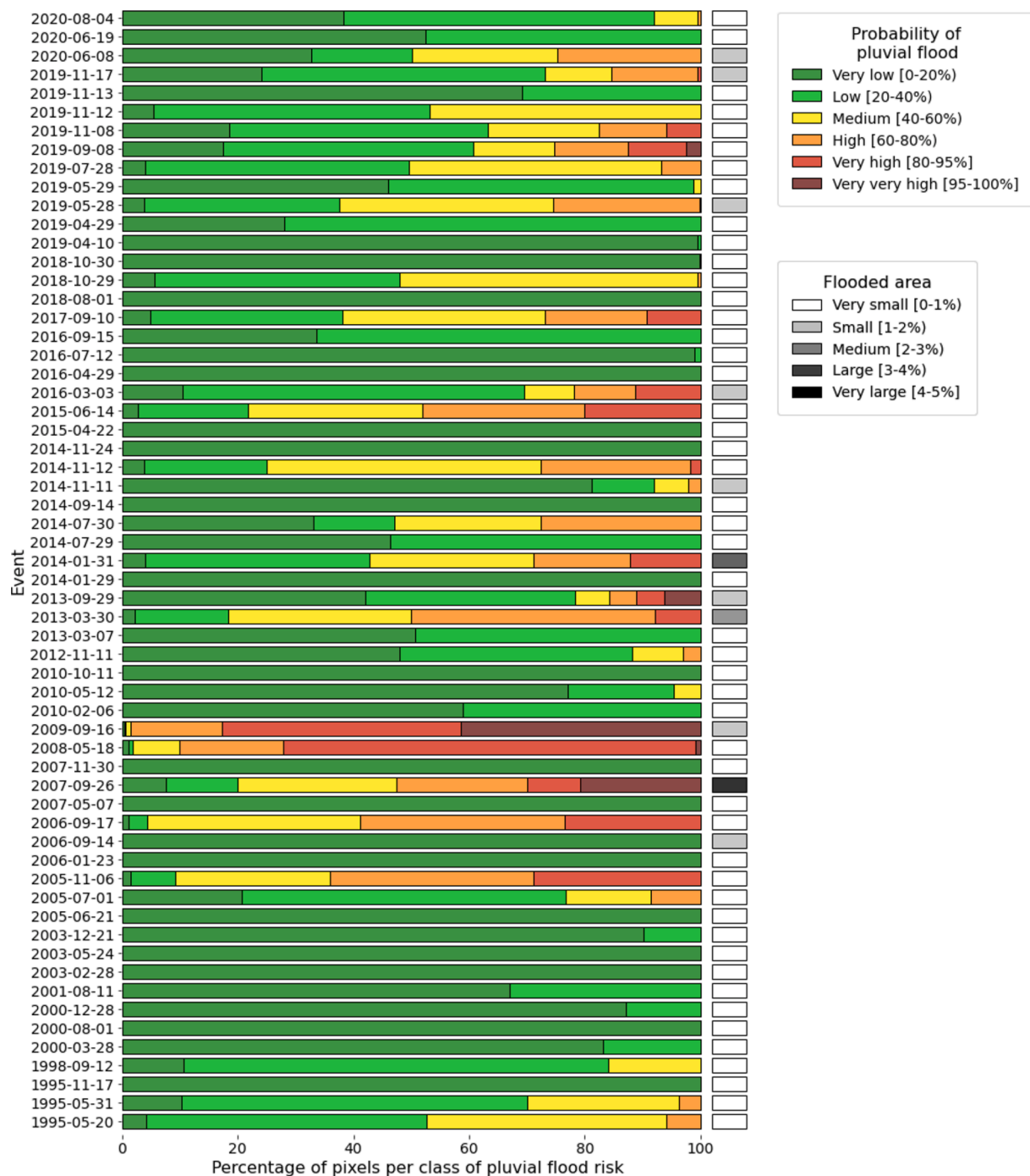


Fig. 5. Percentage distribution of the pluvial flood risk predicted by the model for the investigated 60 events, and comparison with the percentage of the observed flooded areas (in gray chrome).

areas registered (only 26) was observed. Also in this case the mismatch could be explained by an error in reporting the flooded areas, since they have been registered on the day before (November 11th, 2014), where there was a low probability of flooding and low precipitation.

A further evaluation can be done considering Fig. 6, that shows the spatial distribution of the model's predictions and the 6 historical events with the largest pluvial flood areas reported. Also, spatially the model proved to have a high accuracy. The model can identify the most vulnerable areas, showing a good match between the spatial localization of the predicted areas and the historical events.

From Fig. 6 it may appear that the model's predictions overestimate the risk of pluvial flood, as the number of pixels with an associated high

probability of pluvial flood occurrence are always greater than the historical flooded areas. Predicting the areas in which a pluvial flood will occur with high precision is a hard task, but by primarily considering the areas classified to be at very high risk of pluvial flood might help to prioritize the areas more exposed to these events. For example, on September 26th, 2007 (Fig. 6) all the historical flooded areas fall within the highest risk class assessed by the model, with a class probability of 95% – 100% of pluvial flood occurrence. This example shows the importance and usefulness of identifying the areas more exposed according to the estimated risk.

Overall, according to the results (both Figs. 5 and 6), the model tends to overestimate the risk of pluvial flood, even though the largest and

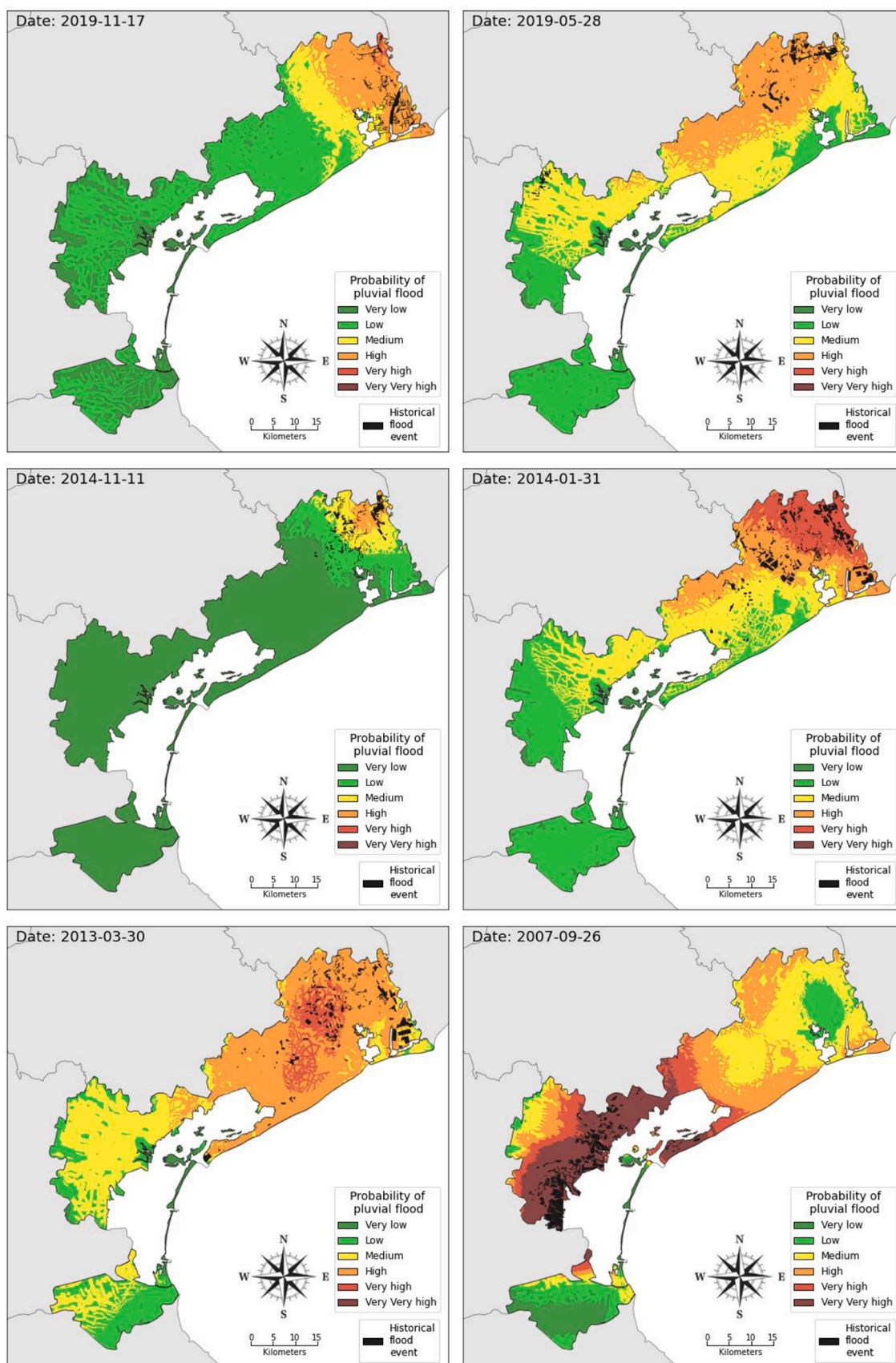


Fig. 6. Comparison between model predictions and pluvial flood areas for the 6 historical events with the largest pluvial flood areas reported.

affecting events were properly recognized by the model. This suggests that the model is accurate and might be applied also for risk scenarios analysis.

At the present state of the authors' knowledge, no other studies applying ML tools for pluvial flood mapping are available in the case study. However, pluvial flood risk and vulnerability maps developed using Multi-Criteria-Decision-Analysis and GIS-based indicators has been provided in Maragno et al. (2020) and Sperotto et al. (2016) for both the entire Metropolitan City of Venice and with a special focus on the Venice Municipality. Specifically, Maragno et al. (2020) provided pluvial flood risk scenarios, based on different rainfall volumes (i.e., 50 and 150 mm of rain), to analyze how different types of land use affect vulnerability to flooding. Sperotto et al. (2016), instead, provided a spatial analysis of areas that are more likely to be at risk of flooding in the future considering projected precipitation for 2041–2050 period and specific territory features (i.e., slope, soil permeability, occurrence of critical hydraulic conditions in the past). Despite, results from these studies cannot be considered fully comparable with the ML approach proposed here, some similar conclusions can be drawn.

Extremely vulnerable areas identified in Maragno et al. (2020) and Sperotto et al. (2016) under worse precipitation scenarios present a quite good spatial matching with the areas predicted by the ML model (i.e. high, very high and very very high probability of flood) under the six worse historical events reported (Fig. 6).

These areas are represented mainly by industrial- commercial and urban fabric located in the central (i.e. Mestre and Venice urban centre and the industrial plan of Porto Marghera) and in the north-east part (i.e. Portogruaro, San Dona', Bibione) of the Metropolitan city of Venice. Some small portions of agricultural areas at high risk are instead located along the coastline.

5. Conclusions

The use of ML can substantially improve and speed up disaster risk assessment and mapping, helping in overcoming most common bottlenecks of physically-based simulations such as the computational complexity and time, the need of large datasets of high-resolution information.

However, most of the standard ML procedures that randomly split the dataset in train and test are not suitable to account for the spatio-temporal variation of pluvial flood, as they risk to violate the assumption of independence between train and test leading to overoptimistic models. The methodology presented in this paper represents a first attempt to build machine learning models able to incorporate the spatio-temporal constraints typical of pluvial flood events and thus to improve rapid pluvial flood risk prediction and mapping capabilities. Accordingly, two different types of cross-validation approaches were applied to investigate the performances of three machine learning models built on historical pluvial flood data and a set of triggering factors representative of the territorial characteristic. As the use of all the available triggering factors can lead to overfitting, a forward features selection was also introduced to reduce the model's complexity. Logistic Regression resulted in the most reliable model, displaying lower overfitting and higher performances, providing robust results both in the validation and test set; while daily cumulative precipitation, distance to rivers and distance to road were the most important features in contributing to pluvial flood risk estimation. Results also confirmed that the LLTO-CV is a crucial step that needs to be considered in the selection of the most suitable ML model for spatio-temporal modeling of pluvial flood risk and to avoid overfitting. The developed model demonstrated to be able to properly reproduce, both in spatial and temporal manner, the most severe floods that occurred in the past in the case study despite some mismatches attributable to the low quality and incompleteness of the historical data collected.

However, some limitations have to be acknowledged. The analysis would have surely benefited from a higher spatial resolution of input

data permitting ML models to better catch relationships between the features and the flooded areas. Furthermore, the model was built on the inventory of historical flood events provided by the local fire brigades. Such dataset at the present state is the best information on past pluvial flood events for the Metropolitan City of Venice but however detailed only the extent of the flooded areas and source of such event. While such variable can be considered enough for the purpose of the analysis (i.e. rapid spatial flood mapping), other important variables should be included (i.e. flood depth, flood velocity) to extend the use of the model also for damage assessment and disaster preparation. In this sense, major efforts should be devoted in the collection of pluvial flood events records.

To build up the model, the number of spatio-temporal folds for cross-validation was set up at 16, while a deeper analysis could find optimal numbers of folds that maximize the learning process of ML models. More accurate methodologies to split the dataset in spatio-temporal folds could also lead to avoid folds without any flooded areas samples, as it occurred for some folds of this analysis.

Future work is envisioned toward a comprehensive methodology that provides not only accurate flooding maps, but also a deeper consideration of the interactions among the multiple factors occurring in the study area. Finally, recognizing the role that other threats (e.g., sea level rise, storm surge, urbanization) can play in increasing pluvial flood risk, future research efforts should be oriented towards a multi-risk approach that considers the potential interaction of climate-related hazards and land use change in the case study area.

Overall, the produced outputs represent a preliminary screening of potential risks posed by pluvial flood and for the identification of vulnerable areas to such impacts at local scale. They can be used to mainstream the definition and implementation of the first phases of local climate change adaptation strategies and plans (e.g., problem formulation, identification of vulnerability and risk hotspots, identification of adaptation priorities). In this perspective, they represent a climate and adaptation service useful to inform the implementation of the Sendai Framework for DRR 2015–2030 (UNISDR 2015) as well as the multi-hazard risk assessments compelled by the EU Civil Protection Mechanism.

The present study must be considered a preparatory exercise for further analyses on pluvial flood risk under climate change conditions. The proposed model has been validated demonstrating to be able to reproduce historical events, paving the way for its application for predicting future pluvial flood events in the Metropolitan City of Venice. Indeed, the outputs of Regional Climate Models (RCMs) as well as future land use scenarios could be integrated in the analyses to assess how future changes in precipitation patterns will affect the occurrence and frequency of pluvial flood events. Therefore, the analyses will help to evaluate how urbanization rates influence flooding patterns, and their extent, becoming a useful tool for decision-makers to identify adaptation measures to be incorporated in spatial planning and disaster risk reduction strategies. In this sense, further analyses could be focused on specific receptors of interest (e.g. people, buildings, infrastructures, agricultural fields, cultural sites) according to local actors' perspectives and needs (Bonato et al., 2021). This can be achieved by including the aforementioned receptors within the model, which will predict not only the spatial distribution of flooded areas, but will also assess the presence of receptors and the possible damages and losses arising in consequence of extreme events.

CRedit authorship contribution statement

Zanetti Marco: Conceptualization, Methodology, Data curation, Software, Formal analysis, Writing – original draft. **Allegrì Elena:** Conceptualization, Formal analysis, Investigation, Validation, Writing – original draft. **Sperotto Anna:** Conceptualization, Investigation, Validation, Writing – original draft. **Torresan Silvia:** Supervision. **Critto Andrea:** Supervision, Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2022.128150>.

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